VocalIQ

AI Speaking Coach

CS651 Web Systems · Prof. Lynne Grewe · Cal State East Bay

Team: Angaddeep Singh Gupta · Samuel Paul Chetty · Abhinay Konuri · Lasya Uma Sri Lingala

vocaliq-956080638663.us-central1.run.app

github.com/angadsinghgupta13112002/vocaliq

Project 2

The Problem

Why does public speaking coaching need to change?

😰

75%

of people suffer from public speaking anxiety

💰

$100–$300

per hour for a human speaking coach

⏳

Days

to get feedback after a presentation

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Zero

accessible, instant, personalized tools existed

"You can't improve what you can't measure."

VocalIQ — Your AI Speaking Coach

Record or upload → Full coaching report in under 90 seconds

Record or Upload

Browser-based recording or file upload — no app needed

Gemini 2.5 Flash AI

3-pass pipeline: assess → coach → gesture analysis

Emotion Timeline

Cloud Vision Face Detection — frame-by-frame emotion + eye contact

Gesture Detection

MediaPipe HandLandmarker — 21 landmarks, 8 gesture types

Scored Coaching Report

Content, Delivery, Confidence, Pacing, Engagement — all scored

Google Drive Integration

Pick videos directly from Drive — no download needed

Technology Stack

Built entirely on Google Cloud

Google OAuth 2.0

Sign-in + Drive auth

Gemini 2.5 Flash

3-pass AI coaching

Cloud Vision API

Emotion + eye contact

MediaPipe

Client-side gestures

Cloud Firestore

Sessions + users

Cloud Storage

Video / audio files

Cloud Run

Containerized deployment

GA4 Analytics

Frontend + backend events

React 19 + Node 20

Vite · Express 5 · JWT

System Architecture

Full-stack on Google Cloud — same origin for frontend + backend

Browser

- React 19 + Vite
- JWT in localStorage
- MediaPipe WASM
- Canvas Frame Extract
- GA4 (gtag.js)

JWT

Express API (Cloud Run)

- POST /coaching/analyze
- POST /analyze-from-drive
- GET /coaching/sessions
- GET /auth/google
- GET /photos/videos

Google Cloud

- Gemini 2.5 Flash
- Cloud Vision API
- Cloud Firestore
- Cloud Storage
- GA4 Analytics

2 vCPU / 2 GiB RAM | 540s timeout | Scale to 0 | ADC — no key files

Three-Pass Gemini Coaching Pipeline

Gemini File API — video uploaded once, analyzed three times

1

Pass 1 — Assessment

Input: Video/audio via Gemini File API

✓ Overall score (1–10)
✓ 5 sub-scores with breakdown
✓ Emotion detected vs needed
✓ WPM (words per minute)
✓ Timestamped transcript
✓ Top 3 improvement tips

2

Pass 2 — Deep Coaching

Input: Pass 1 JSON results as context

✓ Detailed tip cards + exercises
✓ Better phrasing examples
✓ Vocal control breakdown
✓ Strengths to keep
✓ 7-day practice plan

3

Pass 3 — Gesture Analysis

Input: Same Gemini File URI (no re-upload)

✓ Every 3s: gesture label
✓ open_hand, pointing, peace
✓ thumbs_up, fist, no_hands
✓ Expressiveness score 0–100
✓ Drive-uploaded videos only

Authentication — Google OAuth 2.0 + JWT

Server-side token exchange + 7-day signed JWT

1. User clicks "Sign in with Google"
2. Server redirects to Google consent screen
3. Google returns auth code to callback URL
4. Server exchanges code → user profile
5. Server upserts user doc in Firestore
6. Server signs + returns JWT (7-day expiry)
7. Client stores JWT in localStorage

🔒 Security Notes

- Access token NOT stored server-side
- JWT verified on every protected route
- Drive tokens stored encrypted in Firestore
- Auto-refresh on 401 (token expiry)

📦 Google Drive OAuth

- Separate OAuth flow (drive.readonly scope)
- Consent shown every time for refresh token
- Token stored in users Firestore doc
- Auto-refreshed before each Drive API call

Database — Google Cloud Firestore

NoSQL document database in Native mode — two collections

👤 users/{uid}

uid	Google sub ID (document key)
displayName	Full name from Google profile
email	Google email address
photoURL	Profile picture URL
provider	"google"
photosAccessToken	Drive OAuth access token
photosRefreshToken	Drive OAuth refresh token
photosConnectedAt	Drive connection timestamp
updatedAt	Last login timestamp

📊 coaching_sessions/{uid}_{ts}

overallScore	1.0–10.0 overall score
scoreBreakdown	5 sub-scores
transcript	Timestamped segments array
improvementTips	Top 3 tips array
detailedTips	Pass 2 tip cards array
vocalControl	Pacing, tone, breath, pause
emotionTimeline	Cloud Vision per-frame array
emotionSummary	Dominant emotion + eye contact
gestureTimeline	MediaPipe/Gemini gestures
gestureSummary	Dominant gesture + score

All reads/writes via Firebase Admin SDK on Express server — React client never touches Firestore directly

Coaching Report — What You Get

5 tabs of AI-powered, data-driven coaching feedback

📊

Overview

- Score ring (1–10)
- 5 sub-score bars
- Top improvement tips
- Strengths to keep

😊

Emotion Timeline

- Cloud Vision frame analysis
- Emotion color bar
- Eye contact %
- Nervous / confident moments

🖐

Gestures

- MediaPipe timeline bar
- Expressiveness score 0–100
- Gesture frequency chart
- Coaching tip by gesture

📝

Transcript

- Timestamped segments
- Issue tags per segment
- Specific fix per segment
- Full speech text

💡

Tips & Coaching

- Pass 2 tip cards + exercises
- Better phrasing examples
- Vocal control (WPM / tone)
- 7-day practice plan

Analytics & Logging

GA4 Measurement ID: G-SN3LE9Y5RT · Cloud Run stdout logging

Frontend Events (gtag.js)

page_view	Every route change
login	Google OAuth success
session_submitted	Analyze click
session_complete	After AI returns
report_tab_viewed	Each tab click

Backend Events (Measurement Protocol)

login	After JWT issued
session_analyzed	After Firestore save (+ score, emotion, source)

Cloud Run Logging

All server events streamed to Cloud Logging via stdout · Prefixes: [coaching] [gemini] [vision] [storage] [photos] [oauth] [analytics]

Deployment — Google Cloud Run

Multi-stage Docker build · Same URL for frontend + backend

🐳 Multi-Stage Dockerfile

Stage 1 — Build

node:20-alpine · npm ci · npm run build
(Vite compiles React → /client/dist)

Stage 2 — Runtime

node:20-alpine · apk add ffmpeg
Copies /dist + /server · node server.js

🚀 Cloud Run Configuration

Memory	**2 GiB**
CPU	**2 vCPU**
Timeout	**540 seconds**
Instances	**0 – 3 (scale to zero)**
Region	**us-central1**

🔐 Application Default Credentials (ADC) — No key file in Docker image

Cloud Run service account has: Firestore · Cloud Storage · Cloud Vision · Gemini API · GA4 Measurement Protocol

Lessons Learned

Six things we discovered building VocalIQ

Express 5 Breaking Change

app.get("*") is invalid in Express 5 — must use app.use() as the SPA catch-all

Cloud Run PORT

Cloud Run reserves PORT env var — cannot override it. Always read process.env.PORT

ADC vs Key Files

Never commit secrets. Cloud Run ADC + service account roles is cleaner and more secure

Vite Bakes API URLs

Vite bakes env vars at build time — need .env.production with correct API_BASE_URL

GA4 Backend Events

Server-side events need Measurement Protocol — gtag.js is browser-only

Two-Pass Prompting

Pass 1 grounds Pass 2 in facts → dramatically better, more specific coaching advice

Thank You!

Questions & Discussion

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